The pair singular series of a polynomial IV: the second spectrum

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Abstract

Part III reduced the Cesàro form of the off-diagonal of the pair singular series of an irreducible quadratic f to "pieces" $\mathcal{P}_u(Y)$, one for each way the two roots of f can agree on a part u of the modulus, and showed that everything except the pieces' "windows" is understood. This paper is about what the pieces actually are. Numerically each piece is $O(\sqrt{Y})$, and $\mathcal{P}_u(Y)/\sqrt{Y}$ oscillates in $\log Y$ at the Laplace eigenvalues of the *even* Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$, with amplitudes governed by the values of the forms at the Heegner points, or their integrals over the closed geodesics, of discriminant D: a second spectrum, next to the zeros of ζ_K that govern the diagonal (part II). We explain this by an explicit formula: the piece is a twisted Walfisz sum whose Dirichlet series is a sum over frequencies of Dirichlet series of Salié sums, which the Kuznetsov formula of half-integral weight continues to the half-line with poles at $\frac{1}{2} \pm it_j$, and the Katok–Sarnak correspondence identifies the residues as periods of Maass forms. We prove the formula for a single piece in Riesz-mean form [[target]], discuss the uniformity in u that the Cesàro conjecture of part I needs, and report the numerical evidence.

1 Introduction

In plain terms. Parts I and II showed that the main part of the fluctuation of prime values of a polynomial is governed by the zeros of a Dedekind zeta function. The remaining part, the "off-diagonal" of part III, turns out to be governed by a second, entirely different spectrum: the Laplace eigenvalues of the modular surface, the same numbers that govern the distribution of the roots of $x^2 \equiv D \pmod{d}$ (Hooley, Bykovskiĭ, Duke–Friedlander–Iwaniec). Only half of them appear, those of the even forms, and the size of each contribution is the value of the corresponding eigenfunction at a special point, or its integral along a special closed curve, determined by the discriminant of the polynomial. We found this in the data before we had the theory; the theory, once written down, predicts exactly the pattern of presence and absence that the data show.

Set-up. Notation as in part III: $f = t^2 + bt + c$, $D = b^2 - 4c$, $\lambda(p) = p/(p - 2\omega(p))$; for squarefree u with $\omega(u) \geq 1$ and $Q_u(x) = u^2x^2 - D$,

$$\mathcal{P}_u(Y) = \sum_{h \leq Y} (Y - h)\big(F_u(Q_u(h)) - \mathbb{E}F_u\big) - \tfrac{1}{2}(\mathbb{E}F_u - 1)Y, \qquad F_u(n) = \sum_{\substack{d \mid n \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \qquad \mathbb{E}F_u = \sum_{d \text{ adm}} \frac{\lambda(d)\omega(d)}{d^2},$$

admissible meaning squarefree, all primes split, coprime to $2Du$. Part III, Theorem A′: $\mathrm{Off}^*_f(H) = c_{\mathrm{off}}(f)H + \sum_{u \leq H^{2/3+\varepsilon}} w(u)\mathcal{W}_u(H/u; \log H) + O(H(\log H)^{1-c})$, and the Cesàro form of Conjecture 1 of part I is equivalent to the windows being $o(H \log H)$.

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Results.

- *The spectral formula for a piece* (Theorem 1, [to be proved]): for fixed f and u, and Riesz means of order $m \geq m_0$,

$$\mathcal{P}_u^{(m)}(Y) = c_u Y + Y^{1/2} \sum_j \left(\alpha_j(u,f) Y^{it_j} + \overline{\alpha_j(u,f)} Y^{-it_j}\right) \gamma_m(t_j) + O(Y^{1/2-\delta}),$$

  the sum over the even Maass cusp forms u_j of $\mathrm{SL}_2(\mathbb{Z})$ with Laplace eigenvalue $\frac{1}{4} + t_j^2$ (and, for $u > 1$, the newforms of the levels dividing u^2), with $\alpha_j(u,f)$ an explicit multiple of the Katok–Sarnak period of u_j at discriminant D, and $\gamma_m(t) \ll |t|^{-m-1}$.

- *Numerical confirmation* (§6): for five quadratics the even parameters explain the variance of $\mathcal{P}_u(Y)/\sqrt{Y}$ at the 92nd–100th percentile among random frequency sets, the odd ones at the 12th–66th; the dominant line $t_1 = 13.7798$ has stable amplitude and phase over two decades of $\log Y$; the two polynomials with exponentially small periods ($D = -8, -163$) show nothing.

- *What the Cesàro conjecture needs* (§7): the sum of the pieces over $u \leq H^{2/3}$ converges to $c_{\mathrm{off}} H + o(H)$ provided the spectral formula holds with an error $O((H/u)^{1/2+\varepsilon} u^A)$, $A < \frac{1}{4}$; the level dependence of the Kuznetsov formula is therefore the whole remaining problem.

2 The piece as a twisted Walfisz sum

[Sawtooth identity $S_u(t) = \sum_d (\lambda(d)/d) \sum_{x \in R_d^{(u)}} (\frac{1}{2} - \{(t-x)/d\})$; Fourier form; Dirichlet series $\sum_k k^{s-2}\Gamma$-factor$\cdot W_k(s)$, $W_k(s) = \sum_d \lambda(d)\rho_k(d) d^{-s}$; comparison with the classical $\sum_{n \leq x} \frac{1}{n}(\frac{1}{2} - \{x/n\}) = O((\log x)^{2/3})$ (Walfisz) and with the diagonal of part I.]

3 Salié sums and the Kuznetsov formula of weight one half

[Weyl sums of quadratic roots as Salié sums (DFI 1995 §2); the dilation as a fractional frequency; Kuznetsov for the theta multiplier on $\Gamma_0(4)$ (Proskurin) and its use by Bykovskiĭ 1984 and DFI 2012; continuation of $W_k(s)$ to $\Re s > \frac{1}{2}$; the admissibility sieve; the twist $\lambda = 1 * \kappa$.]

4 Katok–Sarnak and the parity rule

[Statement of the Katok–Sarnak correspondence; residues as periods; vanishing for odd forms at i, ρ, over conjugate pairs of Heegner points and reflection-symmetric geodesics; the level $4u^2$ and oldforms.]

5 The spectral formula for a piece

Theorem 1 ([target]). *[statement as in the introduction, with the precise hypotheses on u and m_0.]*

6 Numerical evidence

[Table of NOTES-routeR.md §1; figure of $\mathcal{P}_1(Y)/\sqrt{Y}$ for $t^2 - 2$ against the fitted even lines; stability of amplitudes and phases; scripts.]

7 Uniformity in u and the Cesàro conjecture

[The condition $A < \frac{1}{4}$; what the spectral large sieve gives; the windows of part III in spectral terms.]

AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model was used to draft mathematical arguments, to write and run the software behind every number in the paper, to search the literature, and to draft the text; all of this was done in dialogue with the author, who verified the code and the numerical results, reviewed the proofs, and takes full responsibility for the content. The model is not an author. The complete record of the work is in the public repository `https://github.com/gewure/ulamnd` (directory `research/`).

References